Building resilient intelligence networks for Europe: strengthening governance of evidence for pandemic preparedness, lessons from Belgium and beyond

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Background

The COVID-19 pandemic highlighted the importance of intelligence networks that do not simply collect data but enable its effective translation into policy. Responding to this challenge, the BE-PIN (Belgian Pandemic Intelligence Network) project aims to strengthen Belgium's pandemic preparedness by improving the governance of advisory systems and enhancing the use of scientific evidence in policymaking. To support this objective, we have examined how scientific evidence informed decision-making during the COVID-19 crisis in Belgium.

Methods

We conducted a **systematic literature review,** and analysed the reports of advisory groups. We subsequently conducted interviews with key national experts and policymakers. We performed a **qualitative analysis** to identify strengths, gaps and lessons learnt from the advisory processes in place, which could inform the design and operation of more effective pandemic intelligence networks.

Literature review (Scopus, Pubmed, national and international evaluations); n=22

Advisory group reports; n=44

Semi-structured interviews; n=17

Preliminary results

These insights informed the development of a **conceptual framework** to guide the design and evaluation of governance structures for pandemic intelligence networks. The preliminary results found several **enablers** of effective governance of scientific advisory processes. Conversely, we found that **persistent barriers** impeded the timely and coherent use of evidence.

Examples of enablers	Examples of barriers
• Clearly defined legal mandates	• Fragmented coordination → duplication of work
• Formalised roles and responsibilities between experts and policymakers	• Limited integration of diverse disciplines (e.g., social sciences)
• Advisory body independence	• Lack of structural resourcing
• Transparent expert selection procedure	• Lack of institutional memory → creation of ad hoc bodies

Next steps

To **identify transferable practices** and **refine our recommendations**, this research is being extended to seven additional countries: Austria, Germany, Italy, the Netherlands, Sweden, Switzerland and the United Kingdom using the same methodology. Finally, **focus group discussions** will be organised with key Belgian stakeholders to develop commonly agreed recommendations to strengthen scientific advisory processes for future health crises. The results of this work will also contribute to the upcoming activities of the European Pandemic Preparedness Partnership **BE READY NOW**.

Conclusions

Building resilient, trusted, and responsive health systems requires long-term investment in the governance of evidence-to-policy processes. Intelligence networks, grounded in interdisciplinary collaboration and supported by structured coordination, should not only be activated in times of crisis but be a core function of everyday public health endeavours in order to ensure institutional memory for future crises.

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